

Igor Babin

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Profile

Machine Learning Engineer with 6+ years of experience, including 2+ years as an ML Lead, specializing in Computer Vision, MLOps and Embedded AI. Passionate about transforming cutting-edge AI research into real-world products that reduce costs, improve efficiency, and create new customer experiences.

Education

Master's degree, with honors — **Ukrainian Catholic University**, Computer Science (Data Science) — Lviv, Ukraine, 2022 – 2024
- Thesis topic: Image inpainting through refinement of features space

Bachelor's degree — **National University of Kyiv-Mohyla Academy**, Applied Mathematics — Kyiv, Ukraine, 2017 – 2021

Experience

NDA (military, DefTech), Senior Machine Learning / Computer Vision Engineer — Kyiv, Ukraine, June 2025 – present

End-to-end computer vision solutions — from data collection and annotation preparation to model training, experimentation, and deployment on embedded devices with limited computational resources.
- Proficient in Python, PyTorch, and MLflow for model development and optimization workflows
- Real-time deployment on hardware such as Raspberry Pi 5, Pi Zero, and Hailo NPUs
- Object detection, tracking, and visual navigation tasks
- Model compression techniques (distillation, pruning, quantization) to achieve real-time performance while maintaining robust accuracy

ADVA Soft, Machine Learning Team Lead — Lviv, Ukraine, June 2023 – June 2025

Leads a team of machine learning engineers in developing and implementing machine learning models and systems. Provides technical leadership, manages project timelines, and ensures the successful delivery of ML-powered solutions on devices through the CoreML framework.
- Lead a team of machine learning engineers, providing technical guidance and support
- Collaborate with other teams to help integrate ML solutions into projects
- Collaborate with product owners to define project requirements and goals
- Design and implement machine learning models and algorithms to evaluate and validate ML models' performance and accuracy
- Stay updated with the latest ML algorithms, frameworks, and techniques advancements

ADVA Soft, Machine Learning Engineer | Computer Vision — Lviv, Ukraine, Mar 2021 – June 2023

Develop and implement machine learning models. Work closely with software engineers to build and deploy ML-powered solutions that address business challenges and enhance product functionality.

- Develop and implement machine learning models and algorithms
- Integrate ML models into software systems and applications
- Optimize and fine-tune ML models for improved performance and accuracy

AI Meetups, Lecturer — Lviv, Ukraine — Oct 2023 – June 2024

- Delivering lectures on various AI-related topics
- Searching for speakers in different domains
- Managing technical materials of other speakers

mentor.sh, Mentor and Consultant — Kyiv, Ukraine — July 2025 – present

Binary Studio, Software Engineer — Kyiv, Ukraine — June 2019 – Sept 2020

Volunteer

Community meetups and hackathons, Speaker — Ukraine — Jan 2024 – present

Seasonally delivering lectures on Computer Vision, Machine Learning, and AI topics at various meetups and hackathons, including Ukrainian Catholic University, AI House, AI meetups, and Kyiv AI Day.

Conferences, Conference Reviewer — Jan 2026 – present

Serve as a conference reviewer evaluating paper quality for CVPR 2026 and ECCV 2026 submissions.

Awards

- IYMC Gold Honour — International Youth Math Challenge

Skills

Machine Learning / Computer Vision: Object Detection, Tracking, Visual Navigation, Image Inpainting, Video Object Segmentation, Distillation, Pruning, Quantization

Embedded AI: Raspberry Pi 5, Pi Zero, Hailo NPU, CoreML, Real-time Deployment, Model Compression

Frameworks & Tools: Python, PyTorch, MLflow, MLOps, Experimentation Workflows

Leadership: Team Leadership, Generative AI, Mentoring, Lecturing

Languages

Ukrainian: Native or bilingual

English: Full professional proficiency

Certificates

TensorFlow: Advanced Techniques: DeepLearning.AI

MLOps Mini-course: AI HOUSE

TensorFlow Developer Certificate: Google

Deep Learning Specialization: DeepLearning.AI

Strategic Leadership:

Publications

DeVOS: Flow-Guided Deformable Transformer for Video Object Segmentation — Jan 2024

Igor Babin
IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)

Global Motion Understanding in Large-Scale Video Object Segmentation June 2023
Igor Babin
Conference on Computer Vision and Pattern Recognition (CVPR) Workshops

Interests

Computer Vision: Video Object Segmentation, Image Inpainting, Generative Computer Vision

Embedded AI: Edge Deployment, Model Compression, Real-time Inference